GRADE 11 / Mathematical Literacy / English

Compare tariffs fairly

01 Understand

Learning goals: Calculate a bill with a fixed and variable part • Compare two tariffs • Interpret a break-even point in context

A tariff may have a fixed fee plus a charge per unit. Write down both parts before calculating.

Use the same usage and billing period for each option. A cheaper per-unit rate may still cost more at low usage.

A mathematical break-even value is a comparison tool; actual contracts may also have limits, taxes and once-off fees.

Illustrative data tariffs — not network offers

Usage (GB)	A: R50 + R10/GB	B: R20/GB
2	R70	R40
5	R100	R100
8	R130	R160

02 Follow the reasoning

Which plan costs less at 8 GB? At what usage do the bills match?

1. A costs 50 + 10 × 8 = R130. B costs 20 × 8 = R160.

2. At break-even, 50 + 10u = 20u, so u = 5 GB and both cost R100.

3. Below 5 GB B is cheaper; above 5 GB A is cheaper, under these simplified tariff rules.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Calculate A at 12 GB.

2. How much is saved using A instead of B at 12 GB?

3. Why should a buyer still inspect the full contract?

04 Check and correct

1. R50 + R120 = R170.

2. B costs R240, so the saving is R70.

3. Possible fair-use limits, coverage, fees, expiry and contract terms may affect the decision.

Common mistake to avoid

Do not multiply the fixed fee by the number of units.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Finance: tariffs and financial documents

DBE CAPS, Grades 10–12: Mathematical Literacy. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID matlit-11.